

Student Name	Balaji MJ
Register Number	TU-6253210211004
Task Name	Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator
Week	Week 5
Subject Area	Business Analytics (BA)

WEEK-5 PROJECT

Task Name:

Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Description:

Students must design a multi-asset financial risk simulator. They are required to compute the historical covariance matrix for 500 assets, execute a Cholesky decomposition to correlate multi-dimensional random noise, simulate independent geometric Brownian motion paths, and calculate the portfolio's 99% Value-at-Risk (VaR).

The project combines portfolio analytics, multivariate statistics, numerical linear algebra, stochastic simulation, and risk measurement. The implementation should show the complete process from historical return data through correlated random shocks, simulated asset prices, portfolio profit-and-loss distribution, and the final 99% VaR estimate.

1. PROJECT OBJECTIVE

The main objective is to build a multi-asset Monte Carlo simulator capable of modelling correlated market movements and estimating portfolio downside risk. The project should demonstrate how covariance structure and stochastic price simulation affect the distribution of future portfolio values.

2. CORE REQUIREMENTS

- Work with a universe of 500 assets.
- Compute historical returns and the 500×500 covariance matrix.
- Perform Cholesky decomposition of the covariance matrix.
- Generate multi-dimensional random noise and correlate it using the Cholesky factor.
- Simulate asset price paths using Geometric Brownian Motion (GBM).
- Construct simulated portfolio values and P&L; outcomes.
- Calculate the 99% Value-at-Risk from the simulated loss distribution.

3. INPUT DATA AND PORTFOLIO SETUP

The simulator requires historical price or return data for 500 assets together with the portfolio allocation. The historical period and data frequency should be documented in the final project.

Recommended input fields

- Date or timestamp
- Asset identifier
- Historical closing price or adjusted price
- Portfolio weight or position value
- Optional asset sector or category

Historical return calculation

For simple returns, the historical return of asset i at time t can be calculated from consecutive prices:

$$r(i,t) = P(i,t) / P(i,t-1) - 1$$

The return matrix contains one column for each asset and one row for each historical observation. For 500 assets, the covariance matrix will contain 500 rows and 500 columns.

Portfolio weights

Let w represent the portfolio-weight vector. The weights can represent capital allocations across the 500 assets. The project should document whether weights are long-only, allow short positions, or use another portfolio construction rule.

$$w = [w_1, w_2, \dots, w_{500}]$$

Illustrative portfolio configuration

Parameter	Illustrative value
Number of assets	500
Historical observations	Project-defined
Simulation model	Geometric Brownian Motion
Simulation method	Monte Carlo
Risk confidence level	99% VaR

4. HISTORICAL COVARIANCE MATRIX

The covariance matrix measures how the returns of the 500 assets move together. It is central to the project because independent random shocks cannot reproduce realistic cross-asset relationships.

Covariance matrix

Let R be the matrix of historical asset returns. The covariance matrix can be represented as:

$$\Sigma = \text{Cov}(R)$$

For 500 assets, Σ is a 500×500 matrix. The diagonal contains individual asset return variances, while off-diagonal entries represent pairwise covariances.

Interpretation

- Positive covariance indicates that two asset returns tend to move in the same direction.
- Negative covariance indicates that two asset returns tend to move in opposite directions.
- Values near zero indicate weaker linear co-movement.
- The diagonal elements represent variances of individual assets.

Example matrix structure

	Asset 1	Asset 2	Asset 3	...	Asset 500
Asset 1	Var■	Cov■■	Cov■■	...	Cov■,500
Asset 2	Cov■■	Var■	Cov■■	...	Cov■,500
Asset 3	Cov■■	Cov■■	Var■	...	Cov■,500
■	■	■	■	■	■
Asset 500	Cov■■■■,■	Cov■■■■,■	Cov■■■■,■	...	Var■■■■

Validation

- Confirm the covariance matrix has shape 500×500 .
- Check that the matrix is symmetric within numerical tolerance.
- Check for missing or invalid return values before decomposition.

5. CHOLESKY DECOMPOSITION

Cholesky decomposition converts a suitable symmetric positive-definite covariance matrix into a lower-triangular matrix. This factor is used to transform independent standard normal random variables into correlated shocks.

Mathematical form

$$\Sigma = L L^T$$

Here Σ is the covariance matrix and L is the lower-triangular Cholesky factor.

Why Cholesky decomposition is required

- Generate independent standard normal noise Z .
- Transform Z using the Cholesky factor.
- Obtain random shocks with the desired covariance structure.
- Preserve historical relationships between the 500 asset return processes in the simulation.

Correlated random shocks

$$\varepsilon = L Z$$

If Z contains independent standard normal variables, multiplying by L creates correlated shocks consistent with the covariance structure.

Implementation considerations

- Use a numerically stable covariance estimate.
- Check whether the covariance matrix is positive definite before decomposition.
- If decomposition fails because of numerical issues, document the matrix-conditioning procedure used.
- Keep the asset ordering identical between the covariance matrix, Cholesky factor, weights, and simulation arrays.

6. GEOMETRIC BROWNIAN MOTION SIMULATION

Each asset can be simulated using Geometric Brownian Motion. The model represents continuous stochastic price evolution using a drift component and a random shock component.

GBM equation

$$dS(t) = \mu S(t) dt + \sigma S(t) dw(t)$$

For discrete simulation over a time step Δt , the commonly used exact discretisation is:

$$S(t+\Delta t) = S(t) \times \exp[(\mu - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t} Z]$$

In the multi-asset version, the independent Z values are replaced by correlated shocks produced through the Cholesky factor.

Simulation workflow for 500 assets

- Set initial price S_0 for every asset.
- Estimate historical drift parameters or use the project-defined assumptions.
- Estimate volatility from historical returns.
- Generate independent standard normal random variables.
- Apply Cholesky correlation to create multi-asset shocks.
- Update all 500 asset prices for each simulation time step.
- Repeat until the selected horizon is reached.
- Repeat the complete path for many Monte Carlo scenarios.

Illustrative parameters

Parameter	Example
Assets	500
Time horizon	1 day / project-defined
Time step	Daily or project-defined
Monte Carlo paths	10,000 or project-defined
Confidence level	99%

7. PORTFOLIO VALUE AND PROFIT-AND-LOSS SIMULATION

After simulating future prices for all 500 assets, the model converts those asset-level outcomes into portfolio values.

Portfolio value

$$V_t = \sum_{i=1}^N w_i S_{i,t}$$

At the simulation horizon, calculate the corresponding simulated portfolio value V_T for every Monte Carlo path.

Portfolio profit and loss

$$P\&L = V_T - V_0$$

For a loss-based VaR calculation, the negative side of the simulated P&L; distribution is the relevant risk region.

Monte Carlo workflow

- Generate one correlated multi-asset shock sequence.
- Simulate the 500 asset prices to the selected horizon.
- Calculate the portfolio's terminal value.
- Calculate P&L; relative to the initial portfolio value.
- Store the P&L; for that simulation path.
- Repeat for all Monte Carlo paths.

Illustrative simulation output

Simulation	Terminal Portfolio	P&L;
1	■9,92,500	-■7,500
2	■10,21,000	+■21,000
3	■9,75,500	-■24,500
...	...	...
10,000	■9,84,000	-■16,000

The numerical values above are illustrative only. The final project must calculate terminal values and P&L; from the actual simulation.

8. CALCULATE 99% VALUE-at-RISK (VaR)

Value-at-Risk summarises a loss threshold associated with a selected confidence level. At the 99% level, the VaR is derived from the lower tail of the simulated P&L; distribution.

Historical / Monte Carlo percentile approach

For simulated P&L; values, determine the 1st percentile because the 99% confidence VaR focuses on the lower 1% tail of losses.

$$\text{VaR}_{99\%} = -\text{Quantile}(\text{P\&L}, 0.01)$$

The sign convention should be stated clearly in the report. Under this convention, a positive VaR represents the magnitude of the loss threshold.

Illustrative example

Suppose the 1st percentile of simulated P&L; is $-\text{£}85,000$. Then:

$$\text{VaR}_{99\%} = -(-\text{£}85,000) = \text{£}85,000$$

This means the simulated 1% lower-tail P&L; boundary corresponds to a loss magnitude of $\text{£}85,000$ under the model assumptions.

VaR interpretation

- A higher VaR magnitude indicates a larger modelled downside threshold for the selected horizon.
- VaR depends on the portfolio, historical inputs, simulation assumptions, time horizon, and confidence level.
- Monte Carlo VaR is a model-based estimate rather than a guarantee of future losses.

Optional risk diagnostics

- Plot the simulated P&L; distribution.
- Mark the 1st percentile on the distribution.
- Report the worst simulated losses.
- Repeat the simulation with different random seeds to examine stability.

9. COMPLETE SYSTEM ARCHITECTURE

The complete project should connect historical data, covariance estimation, Cholesky correlation, GBM simulation, portfolio valuation, and VaR calculation.

Historical Prices for 500 Assets
↓
Data Cleaning and Return Calculation
↓
500 × 500 Covariance Matrix
↓
Cholesky Decomposition: $\Sigma = LL^T$
↓
Independent Standard Normal Noise
↓
Correlated Multi-Asset Shocks
↓
Geometric Brownian Motion
↓
Monte Carlo Price Paths
↓
Portfolio Terminal Values
↓
Simulated P&L; Distribution
↓
1st Percentile / 99% VaR
↓
Risk Report and Visualizations

10. RECOMMENDED TECHNOLOGIES

- Python as the primary programming language.
- NumPy for matrix operations and random-number generation.
- Pandas or Polars for historical market data processing.
- SciPy for numerical routines and validation where appropriate.
- Matplotlib or Plotly for risk and simulation visualizations.

Implementation requirement

The core project should demonstrate the covariance, Cholesky, correlated-noise, GBM, portfolio-P&L, and VaR calculations. Libraries may support numerical operations, data handling, and validation, but the report should explain the mathematical logic and workflow.

11. PERFORMANCE AND VALIDATION

A 500-asset Monte Carlo simulator can be computationally intensive. The project should document both numerical correctness and execution performance.

Performance metrics

- Number of assets: 500
- Number of historical observations
- Number of Monte Carlo paths
- Number of simulation time steps
- Covariance calculation time
- Cholesky decomposition time
- Monte Carlo simulation time
- Peak or approximate memory usage

Validation checks

- Verify that the return matrix has the expected dimensions.
- Verify that the covariance matrix is 500×500 and symmetric.
- Verify that the Cholesky factor reconstructs the covariance matrix within numerical tolerance.
- Check that correlated shocks have approximately the intended covariance structure over a large sample.
- Verify that simulated prices remain positive under the GBM formulation.
- Check that the P&L; distribution contains both gains and losses where expected.
- Verify that the 99% VaR uses the correct lower-tail percentile and sign convention.

Recommended visualizations

- Heatmap of the 500×500 covariance or correlation matrix.
- Distribution of simulated portfolio P&L;.
- 99% VaR threshold marked on the P&L; distribution.
- Selected simulated asset-price paths.
- Convergence of VaR as the number of Monte Carlo paths increases.

12. FINAL PROJECT DELIVERABLES

1. Historical price/return data preparation for 500 assets.
2. 500×500 historical covariance matrix.
3. Cholesky decomposition and numerical validation.
4. Independent random-noise generation.
5. Correlated multi-dimensional random shocks.
6. Geometric Brownian Motion implementation.
7. Monte Carlo simulation of multi-asset price paths.
8. Portfolio terminal-value and P&L; calculation.
9. 99% Value-at-Risk calculation.
10. Risk distribution visualizations.
11. Performance and memory analysis.
12. Final risk report documenting assumptions and results.

13. PROJECT OBJECTIVE

The main objective of this Week-5 Business Analytics project is to design a multi-asset financial risk simulator that captures cross-asset dependence and estimates portfolio downside risk. The project combines a historical covariance matrix, Cholesky decomposition, correlated random shocks, Geometric Brownian Motion, Monte Carlo simulation, and 99% VaR into one end-to-end analytical workflow.

14. FINAL OUTPUT

Output	Description
Covariance Matrix	500×500 historical asset covariance matrix
Cholesky Factor	Lower-triangular factor used for correlated shocks
Simulated Paths	Multi-asset GBM Monte Carlo price paths
P&L; Distribution	Simulated portfolio profit-and-loss outcomes
99% VaR	Lower-tail portfolio loss threshold
Risk Report	Summary of assumptions, metrics, visualizations, and results

Note: Numerical values shown in examples are illustrative. Final covariance values, simulation outputs, P&L; distribution, and 99% VaR must be calculated from the actual dataset, portfolio weights, simulation configuration, and random seed used in the project.